

Thermodynamics Notes

EGR 3302

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Lecture 2: Thermodynamic Systems and Properties

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0.0.1 Introduction to Thermodynamic State

When we study a system, we do that through investigating its properties. We call the *state* the collection of all of the system's independent properties.

The system will evolve through the space of configuration. It does that through what's called a *process*, and it takes a *path* to reach a certain equilibrium (under ergodic assumptions).

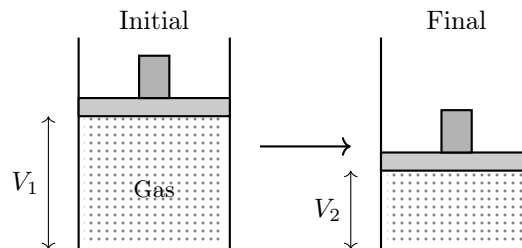


Figure 1: Piston-cylinder system compression process.

Definition 1 (State Postulate). The state of simple compressible substances is fully determined by **two** independent properties.

0.0.2 Types of Processes

Thermodynamic processes can be classified according to which property remains constant. Using the language of partial derivatives, we have:

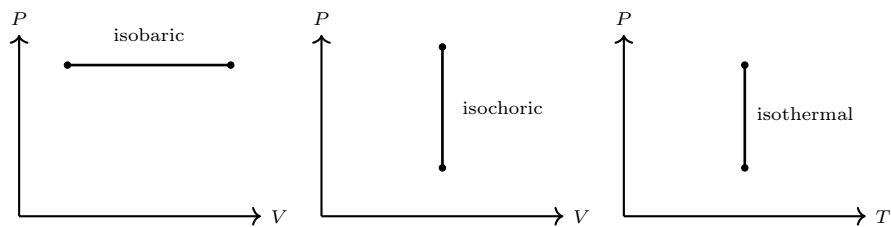


Figure 2: Thermodynamic processes: isobaric and isochoric (P - V), isothermal (P - T) as vertical bar.

Process	Constant
Isobaric	Pressure: $\partial P / \partial t = 0$
Isothermal	Temperature: $\partial T / \partial t = 0$
Isochoric	Volume: $\partial V / \partial t = 0$

Definition 2 (Quasi-Static Process). A **quasi-static process** is one that occurs slowly enough that the system remains in internal equilibrium at each instant. This is an idealization, but useful for analysis.

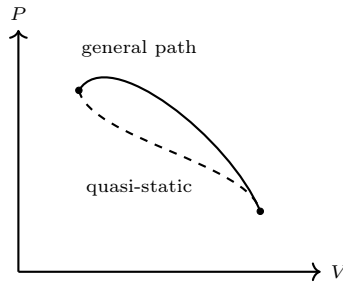


Figure 3: Minimal P - V diagram showing two exaggerated paths between the same initial and final states: quasi-static (solid) and general (dashed).

0.0.3 Key Thermodynamic Properties

The fundamental properties we use to describe thermodynamic systems are:

- **Volume** (V): The space occupied by the system [m^3]
- **Temperature** (T): A measure of thermal energy [K]
- **Pressure** (P): Force per unit area [$\text{Pa} = \text{N}/\text{m}^2$]

Note (Heat vs. Temperature). Heat is an *interaction* (energy transfer), while temperature is a *property* (state variable).

Temperature

Temperature is a measure of the average kinetic energy of molecules in a system in thermal equilibrium.

For **monatomic ideal gases**, the relationship between temperature and molecular kinetic energy is:

$$T = \frac{2}{3k_B} \left\langle \frac{1}{2}mv^2 \right\rangle$$

where $k_B = 1.381 \times 10^{-23} \text{ J/K}$ is Boltzmann's constant, m is the molecular mass, and $\langle v^2 \rangle$ is the mean square velocity.

Alternatively, we can write:

$$\left\langle \frac{1}{2}mv^2 \right\rangle = \frac{3}{2}k_B T$$

Note (Temperature Scales).

$$T_K = T_C + 273.15$$

$$T_C = T_K - 273.15$$

Pressure

Pressure is the normal force per unit area exerted by a fluid:

$$P = \frac{F}{A} \quad [\text{Pa} = \text{N/m}^2]$$

Pressure arises from molecular collisions with the container walls. Each collision transfers momentum, creating a force.

Remark. Atmospheric pressure is quite large: $P_{\text{atm}} \approx 101,325 \text{ kPa} \approx 101 \text{ kPa}$.

0.0.4 Example Problems

Example (Piston-Cylinder System). Consider a piston-cylinder device containing a fluid. The piston has mass m and cross-sectional area A . Find the fluid pressure.

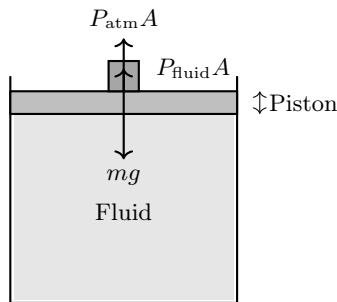


Figure 4: Piston-cylinder system showing force balance on the piston.

Solution: Force balance on the piston in equilibrium:

$$P_{\text{fluid}} \cdot A = P_{\text{atm}} \cdot A + mg$$

Therefore:

$$P_{\text{fluid}} = P_{\text{atm}} + \frac{mg}{A}$$

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Example (Pressure in the Ocean). Consider a fluid element in the ocean. Find how pressure varies with depth.

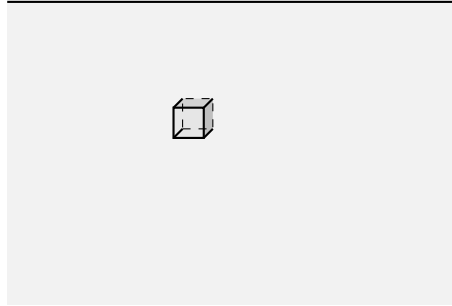


Figure 5: Fluid element in hydrostatic equilibrium.

Solution: Consider a small cube of fluid with sides dx , dy , dz . The vertical force balance gives:

$$P(z) \cdot dx \, dy - P(z + dz) \cdot dx \, dy - \rho g \, dx \, dy \, dz = 0$$

Dividing by $dx \, dy \, dz$ and taking the limit:

$$\frac{\partial P}{\partial z} = -\rho g$$

For constant density ρ , integrating from the surface (where $P = P_{\text{atm}}$) to depth h :

$$P(h) = P_{\text{atm}} + \rho g h$$

where ρ is the fluid density [kg/m^3], $g = 9.81 \, \text{m}/\text{s}^2$ is gravitational acceleration, and h is the depth below the surface [m].

Numerical example: At 10 m depth in water ($\rho \approx 1000 \, \text{kg}/\text{m}^3$):

$$P = 101,325 + (1000)(9.81)(10) \approx 199 \, \text{kPa} \approx 2 P_{\text{atm}}$$

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Lecture 3: Work, Energy, and the First Law

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0.0.5 Applications of Thermodynamics

Thermodynamics is fundamental to the energy generation industry. Most thermal power plants operate on similar principles: fuel is burned to generate heat, which expands a gas, producing mechanical work and ultimately electrical energy.

Beyond power generation, thermodynamics governs the transportation industry from automobile engines to jet propulsion systems.

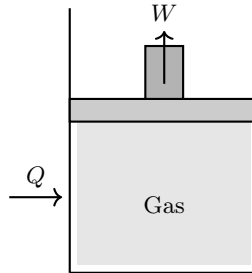
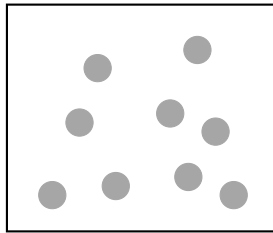


Figure 6: Piston-cylinder system.

0.0.6 Conservation Laws: A Motivating Example

Consider a box containing several identical balls. We can determine the number of balls using:

$$N = \frac{m_{\text{total}} - m_{\text{box}}}{m_{\text{ball}}}$$



System boundary

Figure 7: Box containing balls.

Suppose we measure the number of balls at two different times, t_1 and t_2 . If $N_2 - N_1 = \Delta N = 0$, we have conservation of the number of balls.

However, what if we find $\Delta N = -1$ (one ball is missing)? This suggests we need to account for exchanges across the system boundary:

$$\Delta N = N_{\text{in}} - N_{\text{out}}$$

This leads to the principle: *nothing can be created or destroyed within the system; changes must be accounted for by transfers across the boundary.*

Note (State vs. Path Functions). N is a **state function**: it depends only on the current state. ΔN is a **path function**: it depends on the process.

0.0.7 The First Law of Thermodynamics

Theorem 1 (First Law of Thermodynamics). The change in energy of a system equals the energy exchanged with its surroundings:

$$\Delta E = Q + W$$

where ΔE is the change in internal energy, Q is the heat transferred into the system, and W is the work done on the system.

Note (Sign Convention). $Q > 0$: heat flows into the system. $W > 0$: work is done on the system (compression).

Differential Form

For infinitesimal changes:

$$dE = \delta Q + \delta W$$

Here, dE is an exact differential (state function), while δQ and δW are inexact differentials (path functions).

The rate form:

$$\frac{dE}{dt} = \dot{Q} + \dot{W}$$

0.0.8 Key Insights

Remark (Energy is a State Function). The internal energy E depends only on the current thermodynamic state (defined by T, P, V). It does not depend on how the system reached that state.

Remark (Heat and Work are Path Functions). Both Q and W depend on the specific process path. Different processes connecting the same states will involve different amounts of heat and work, but $Q + W = \Delta E$ remains constant.

Remark (Conservation of Energy). The First Law states: energy cannot be created or destroyed, only converted or transferred.

0.0.9 Properties of Energy

Units and Dimensions

Energy has dimensions of $[M L^2 T^{-2}]$ and is measured in:

- **Joules [J]**: The SI unit of energy. $1 \text{ J} = 1 \text{ N} \cdot \text{m} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^2$
- **Kilojoules [kJ]**: $1 \text{ kJ} = 1000 \text{ J}$

For the total internal energy E of a system:

$$E \quad [\text{J}]$$

Specific Energy

Often it is useful to work with energy per unit mass, called **specific energy**:

$$e = \frac{E}{m} \quad [\text{J/kg}]$$

Similarly, we define specific heat and specific work:

$$q = \frac{Q}{m} \quad [\text{J/kg}], \quad w = \frac{W}{m} \quad [\text{J/kg}]$$

The First Law in specific form:

$$\Delta e = q + w$$

Power

Power is the rate of energy transfer or conversion:

$$P = \frac{dE}{dt} \quad [\text{W}]$$

where the Watt [W] is defined as:

$$1 \text{ W} = 1 \text{ J/s}$$

For heat and work transfer rates:

$$\dot{Q} = \frac{dQ}{dt} \quad [\text{W}], \quad \dot{W} = \frac{dW}{dt} \quad [\text{W}]$$

The First Law in rate form:

$$\frac{dE}{dt} = \dot{Q} + \dot{W}$$

Example (Unit Conversion). A system absorbs 500 kJ of heat over 10 s. What is the average rate of heat transfer?

Solution:

$$\dot{Q} = \frac{Q}{t} = \frac{500 \text{ kJ}}{10 \text{ s}} = 50 \text{ kJ/s} = 50 \text{ kW}$$

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0.0.10 Energy from First Principles

Recall from Physics I that Newton's second law gives:

$$m\ddot{x} = F$$

Multiplying both sides by dx and integrating:

$$m\ddot{x} dx = F dx$$

The left side can be rewritten:

$$m\ddot{x} dx = m \frac{dv}{dt} dx = mv dv = d\left(\frac{1}{2}mv^2\right)$$

This gives us:

$$d\left(\frac{1}{2}mv^2\right) = \delta W$$

where $\delta W = F dx$ is the work done. Thus:

$$dE_k = \delta W$$

For conservative forces, we can write $\delta W = -dE_p$, where E_p is potential energy. This leads to conservation of mechanical energy:

$$E_m = \frac{1}{2}mv^2 + mgz = \text{constant}$$

0.0.11 Total Energy of a Thermodynamic System

In thermodynamics, we extend this concept. The total energy of a system includes both mechanical energy and internal energy:

$$E = me = E_m + U$$

where:

- $E_m = \frac{1}{2}mv^2 + mgz$ is the mechanical energy (kinetic + potential)
- U is the internal energy
- $e = E/m$ is the specific total energy

Expanding fully:

$$E = \frac{1}{2}mv^2 + mgz + U$$

Internal Energy Components

The internal energy U can be further decomposed:

$$U = U_{\text{sensible}} + U_{\text{latent}}$$

- **Sensible energy:** associated with temperature (molecular kinetic energy and vibrations)
- **Latent energy:** associated with phase changes (molecular bond energy)

For most applications in thermodynamics, when the system velocity and elevation are negligible or constant:

$$E \approx U$$

and the First Law simplifies to:

$$\Delta U = Q + W$$

0.0.12 Types of Work

Electric Work

Consider an electrical circuit with a resistor. When current flows through a resistance, electrical work is converted to heat.

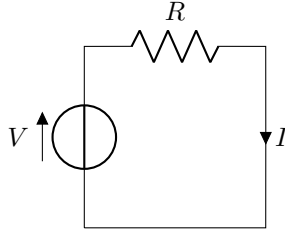


Figure 8: Simple resistive circuit.

The voltage across the resistor is related to the electric field:

$$V = E\ell$$

where E is the electric field and ℓ is the length.

The electrical power (rate of work) is:

$$\dot{W}_{\text{elec}} = VI$$

The work done over a time interval Δt is:

$$W_{\text{elec}} = \dot{W}_{\text{elec}} \Delta t = VI\Delta t$$

Using Ohm's law ($V = IR$), we can also write:

$$\dot{W}_{\text{elec}} = I^2 R = \frac{V^2}{R}$$

Shaft Work

Consider a rotating shaft with a tangential force F applied at radius r .

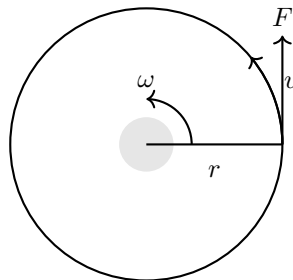


Figure 9: Rotating shaft with tangential force.

The power delivered by the force is:

$$\dot{W}_{\text{shaft}} = \vec{F} \cdot \vec{v}$$

For a tangential force at radius r with angular velocity ω :

$$v = r\omega$$

The torque is:

$$\tau = Fr$$

Therefore, the shaft power is:

$$\dot{W}_{\text{shaft}} = Fv = F(r\omega) = \tau\omega$$

The work done over a time interval Δt is:

$$W_{\text{shaft}} = \dot{W}_{\text{shaft}} \Delta t = \tau\omega\Delta t$$

Or in terms of angular displacement $\theta = \omega\Delta t$:

$$W_{\text{shaft}} = \tau\theta$$

Note (Units). For shaft work:

- Torque: τ [N · m]
- Angular velocity: ω [rad/s]
- Power: $\dot{W}_{\text{shaft}}$ [W]

Lecture 4: Moving Boundary Work

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0.0.13 Decomposing Work

We often express work as a function of the system boundary plus other work:

$$\delta W = \delta W_b + \delta W_{\text{other}}$$

Here δW_b is the **boundary work**, also called **moving boundary work** or **P - V work**, which comes from volume changes at the system boundary.

0.0.14 Piston-Cylinder System

Consider a piston-cylinder system with a gas inside. The piston has mass m and cross-sectional area A . We add heat Q from below, the gas expands, and the piston moves upward doing work W .

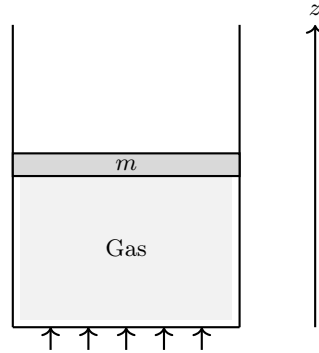


Figure 10: Minimal piston-cylinder system with heat source.

The gas inside exerts pressure P and force $F = PA$ pushing up on the piston. The outside environment exerts pressure P_{ext} and force $F' = P_{\text{ext}}A$ pushing down from above.

0.0.15 Deriving Boundary Work

From mechanics, work is force times displacement. If the piston moves up by dz :

$$\begin{aligned}\delta W_b &= -F dz \\ &= -P_{\text{ext}} A dz \\ &= -P_{\text{ext}} \underbrace{A dz}_{dV} \\ &= -P_{\text{ext}} dV\end{aligned}$$

where $dV = A dz$ is the volume change. The minus sign shows up because when the gas expands ($dV > 0$), work is done *by* the system, giving $\delta W_b < 0$. Recall our sign convention: positive work means work done *on* the system.

Quasi-Static Processes

For a **quasi-static process**, the process happens slowly enough that the system stays in equilibrium. This means:

$$P \approx P_{\text{ext}}$$

So the boundary work becomes:

$$\delta W_b = -P_{\text{ext}} dV \approx -P dV$$

Integrating from state 1 to state 2:

$$W_b = - \int_1^2 P dV$$

Note (Path Dependence). Since W_b is path-dependent, we need $P(V)$ to evaluate this integral. Boundary work is a **path function**, not a state function.

0.0.16 P-V Diagram: Work as Area

On a P - V diagram, boundary work shows up as the area under the curve:

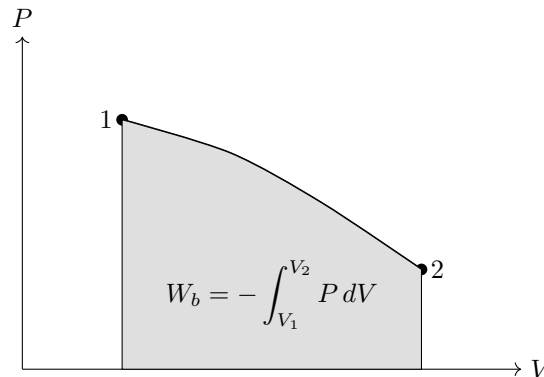


Figure 11: P - V diagram showing boundary work as the area under the process curve.

Theorem 2 (Moving Boundary Work). For a quasi-static process:

$$W_b = - \int_{V_1}^{V_2} P(V) dV$$

0.0.17 Specific Process Types

We'll look at boundary work for a few important process types. Each has its own path on the P - V diagram and its own work expression.

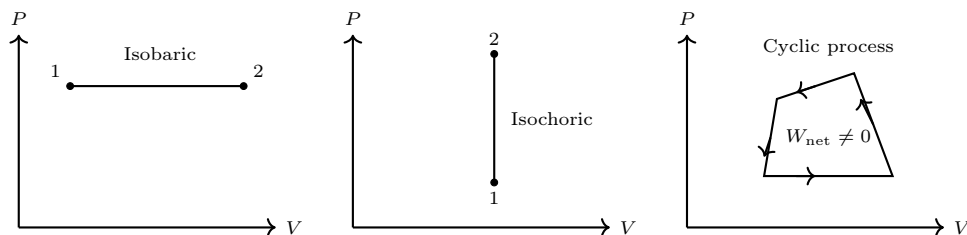


Figure 12: P - V diagrams: isobaric (horizontal line), isochoric (vertical line), and a cyclic process (enclosed area represents net work).

Isobaric Process ($P = \text{const}$)

For constant pressure, the pressure stays fixed while volume changes:

$$W_b = - \int_{V_1}^{V_2} P dV = -P \int_{V_1}^{V_2} dV$$

$$\boxed{W_b = -P(V_2 - V_1)}$$

If $V_2 > V_1$ (expansion), then $W_b < 0$, so work is done by the system. If $V_2 < V_1$ (compression), then $W_b > 0$, meaning work is done on the system. On the P - V diagram, isobaric looks like a horizontal line, and the work is just the rectangle underneath.

Isochoric Process ($V = \text{const}$)

For constant volume, there's no volume change:

$$dV = 0$$

$$\boxed{W_b = 0}$$

No piston motion means no boundary work. On the P - V diagram, isochoric is a vertical line with zero area under it.

Cyclic Process

For a cycle, the system goes through a bunch of processes and returns to where it started. Even though $\Delta V = 0$ for the full cycle, the net work isn't zero:

$$W_b = - \oint P dV$$

The work is the negative of the enclosed area on the P - V diagram:

$$\boxed{W_b = -(\text{Area enclosed by the cycle in the } P\text{-}V \text{ diagram})}$$

For a clockwise cycle, $W_b < 0$ (net work out). For counterclockwise, $W_b > 0$ (net work in). This shows boundary work is path-dependent: same start and end states, but different work depending on the path.

0.0.18 Key Insights

Remark (Path Dependence). Boundary work is path-dependent. Unlike state functions like internal energy, the work depends on the specific path between two states. Take two different paths connecting the same states, you'll get different work values.

Remark (Sign Convention). Remember our convention: $W_b > 0$ means work done on the system (compression), while $W_b < 0$ means work done by the system (expansion). This matches the First Law where positive work increases internal energy.

Remark (Quasi-Static Assumption). The formula $\delta W_b = -P dV$ only works for quasi-static processes where the system stays in equilibrium. If things happen fast, internal and external pressures won't match, and this formula breaks down.

Remark (Geometric Picture). The P - V diagram gives us a nice geometric view: boundary work is the area under the curve. This makes it clear why different paths between the same states give different work.

Lecture 5: Moving Boundary Work

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0.0.19 Energy Change

For a system from state 1 to state 2:

$$\Delta E = E_2 - E_1 = E_{\text{exch}} = Q + W,$$

where

- Q : heat added to the system,
- W : work done *by* the system.

For a fixed-mass system:

$$E = me, \quad e = \frac{v^2}{2} + gz + u, \quad u = u(T, V).$$

0.0.20 Particle Model of Matter

Matter consists of particles in constant motion. Phase depends on:

- Bonding forces,
- Kinetic energy (temperature).

Bonding dominant $\Rightarrow$ solid/liquid, kinetic dominant $\Rightarrow$ gas.

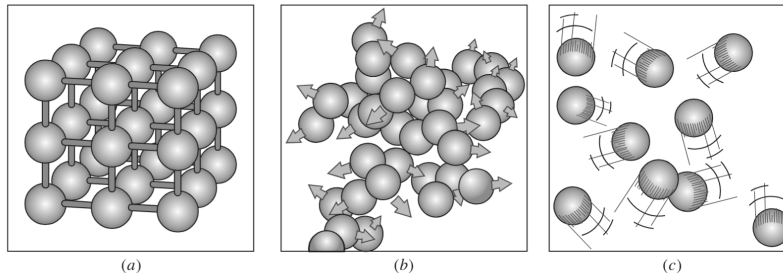


Figure 13: Particle arrangements in the three phases of matter.

Solid

Particles vibrate in a lattice; increasing T eventually breaks bonds $\rightarrow$ melting.

Liquid

Particles close but mobile; moderate bonds allow flow.

Gas

Particles widely separated; negligible interactions except brief collisions.

0.0.21 Phase Changes

Heating a solid:

1. Vibrations increase,
2. Bonds overcome,
3. Bonds break,
4. Phase transition occurs.

Phase Change	From $\rightarrow$ To	Energy
Melting	Solid $\rightarrow$ Liquid	Absorbed
Vaporization	Liquid $\rightarrow$ Gas	Absorbed
Condensation	Gas $\rightarrow$ Liquid	Released
Freezing	Liquid $\rightarrow$ Solid	Released

Note. Temperature constant during phase change; energy breaks/forms bonds.

0.0.22 T - v Diagrams

Specific volume: $v = V/m$. Shows thermodynamic changes.

Heating Water at Constant Pressure

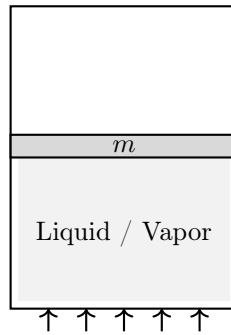


Figure 14: Piston-cylinder system with heat input.

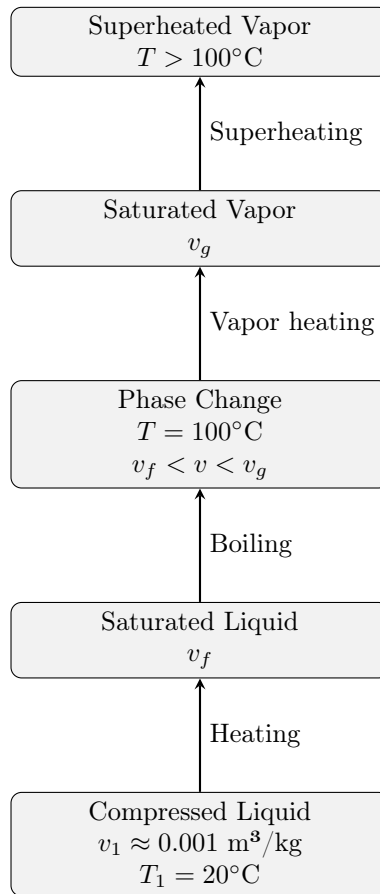


Figure 15: Vertical timeline of water heating at 1 atm (bottom: compressed liquid, top: superheated vapor).

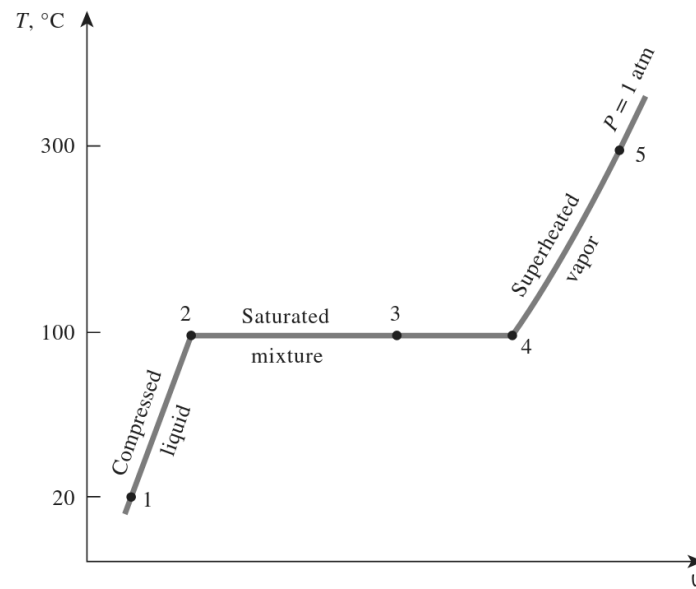


Figure 16: T - v diagram at constant pressure. Horizontal segment: phase change.

Remark (Key Features). Stuff to note down,

- Liquid: T rises, $v \approx \text{constant}$
- Two-phase: T constant, v rises sharply
- Vapor: T and v increase

0.0.23 The T - v Diagram at Various Pressures

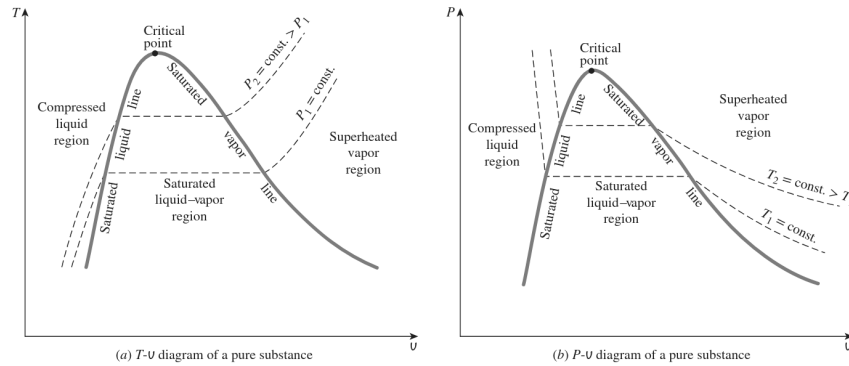


Figure 17: Property diagrams of a pure substance (two-phase domes).

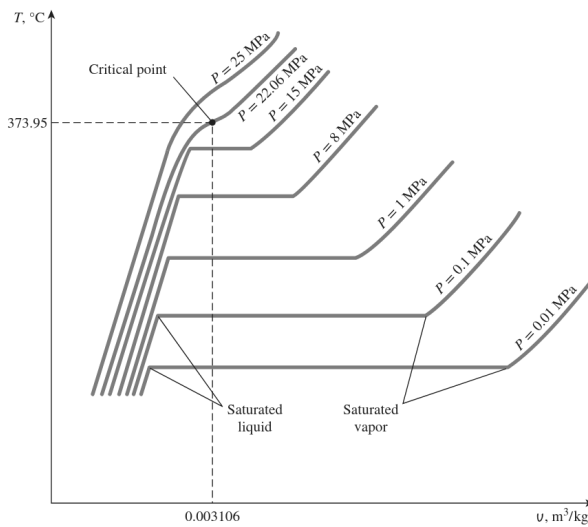


Figure 18: Saturation dome (T - v diagram) showing isobars at various pressures. Dome boundaries meet at the critical point.

Heating at different pressures:

- Boiling temperature rises with pressure (e.g., water: 100°C at 1 atm, 179.9°C at 1 MPa)
- Saturation line shrinks: v_f slightly decreases, v_g decreases more

- Critical point: $P_{\text{cr}} = 22,06 \text{ MPa}$, $T_{\text{cr}} = 373,95 \text{ °C}$, $v_{\text{cr}} = 0,003 106 \text{ m}^3/\text{kg}$

Regions

- Left: Compressed liquid
- Middle: Two-phase mixture
- Right: Superheated vapor